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Leveraging Unsupervised Task Adaptation and Semi-Supervised Learning With Semantic-Enriched Representations for Online Sexism Detection

Francisco Rodríguez-Sánchez[1] | Jorge Carrillo-de-Albornoz[1,2] | Laura Plaza[1,2]

[1]UNED NLP & IR Group, Madrid, Spain | [2]RMIT University, Melbourne, Victoria, Australia

Correspondence: Francisco Rodríguez-Sánchez (frodriguez.sanchez@invi.uned.es)

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ABSTRACT

Over the past decade, the proliferation of hateful and sexist content targeting women on social media has become a concerning issue, adversely affecting women's lives and freedom of expression. Previous efforts to detect online sexism have utilized monolingual ensemble transformers combined with data augmentation techniques that incorporate related-domain data, such as hate speech. However, these approaches often struggle to capture the full diversity and complexity of sexism due to limitations in the size and quality of training data. In this study, we introduce a novel sexism detection system that employs in-domain unlabeled data through unsupervised task-adaptation techniques and semi-supervised learning, using an efficient single multilingual transformer model. Additionally, we incorporate a Sentence-BERT layer to enhance our system with semantically meaningful sentence embeddings. Our proposed system outperforms existing state-of-the-art methods across all tasks and datasets, demonstrating its effectiveness in detecting and addressing sexism in social media text. These results underscore the potential of our approach, providing a foundation for further research and practical applications.

1 | Introduction

The rapid growth of the Internet and digital platforms has introduced new methods of communication and interaction. However, this amplified connectivity also facilitates the spread of harmful and prejudiced behaviors, such as sexism. The Oxford English Dictionary defines sexism as "prejudice, stereotyping or discrimination, typically against women, on the basis of sex." According to Jirovsky (2022), the consequences of online violence are fundamentally the same as those of offline gender-based violence against women, with certain aspects of online violence potentially making it even more severe. In online environments, as in offline settings, sexism encompasses a wide range of behaviors, from overtly hateful expressions to subtle, everyday actions that may be overlooked. Various forms of sexism, such as benevolent, hostile, and ambivalent, can be defined (Glick and Fiske 1996). Regardless of their form, these behaviors have significant negative effects on both individuals targeted and society as a whole (Barreto and Doyle 2023).

Online sexism is a dynamic phenomenon that is constantly changing and adapting. One study by Dickel and Evolvi (2022) examines the framing and discourse surrounding the #MeToo movement within the manosphere, a collection of online communities, including the Men's Rights Movement and Pick-Up Artists, that are often characterized by opposition to feminism and advocacy for traditional gender roles. The researchers identified several recurrent themes in their analysis, including (1) criticism and verbal abuse directed at women, (2) downplaying allegations of rape and presenting the #MeToo movement as a feminist conspiracy, and (3) portraying men as victims while aiming to reestablish patriarchal norms. This study highlights the importance of recognizing the various manifestations of online sexism and exploring how particular communities propagate and reinforce these harmful attitudes. To create a more fair and inclusive online environment, it is essential to address these concerns and devise methods for detecting all forms of sexism. Furthermore, Flouli and Athanasiades (2023) shows how men persist in perpetuating false ideas about female oppression, offering new insights into how male discourse is shaped to maintain male privilege. This illustrates how sexism quickly adapts to different environments and historical contexts.

Historically, the term "sexism" has often been confused with "misogyny" (Fontanella et al. 2024). However, it is important to clarify that they are not the same. Misogyny involves hatred towards women, implying a hateful attitude. Sexism, on the other hand, does not always imply misogyny. Sexism represents a broader concept, encompassing attitudes, beliefs, and behaviors that reinforce stereotypes and inequalities based on gender. Misogyny, in contrast, is a specific form of hateful sexism targeting women and girls, displaying hatred towards them. Consequently, misogyny is a subset of sexism, and while all instances of misogyny are sexist, not all instances of sexism are misogynistic.

In the current research landscape, sexism in online media primarily focuses on detecting misogyny (Fersini, Rosso, and Anzovino 2018; Pamungkas, Basile, and Patti 2020; Guest et al. 2021). Prior state-of-the-art works addressing misogyny detection and categorization into subtypes (Fersini, Rosso, and Anzovino 2018) have mainly employed traditional supervised machine learning approaches trained on various textual features, such as unigrams and bigrams, combined with stylistic and structural features (Pamungkas et al. 2018; Canós 2018; Nina-Alcocer 2018).

Building on the distinction between sexism and misogyny, the EXIST 2021 and 2022 shared tasks (Rodrguez-Sánchez et al. 2021; Rodrguez-Sánchez et al. 2022) focused on identifying and categorizing sexism in online texts. Participants were required first to identify whether a text is sexist or non-sexist (Task 1) and then classify any identified sexist content into specific subcategories (Task 2). Most approaches employed supervised methods utilizing ensembles of multiple transformer models, achieving state-of-the-art results for both tasks (de Paula, da Silva, and Schlicht 2021; Vaca-Serrano 2022). This approach makes the size and quality of the labeled data essential factors. However, training and deploying multiple transformer architectures can be inefficient and challenging, and acquiring labeled data is often limited, expensive, and time-consuming.

To address these issues, data augmentation techniques can be employed to expand and diversify the dataset. For instance, previous studies in domains other than sexism detection have demonstrated that transformer models can benefit from pre-training with Supporting Information (Gururangan et al. 2020). Nonetheless, attempts at detecting sexism have often depended on data from related domains like hate speech (Mina et al. 2021), which might not sufficiently represent the subtleties of sexism as a more comprehensive concept. Other approaches, such as back-translation of training data, have demonstrated success in past research obtaining state-of-the-art results in EXIST 2021 (de Paula, da Silva, and Schlicht 2021). However, these techniques are largely dependent on the amount of labeled data available. Although semi-supervised learning can help mitigate this issue, its application in sexism detection has utilized self-training techniques (Alsafari and Sadaoui 2021), which can be computationally demanding, particularly when working with complex models or large datasets.

In this paper, we propose an unsupervised task-adaptation and semi-supervised approach with semantic-enriched information for sexism identification and categorization into different subtypes. Our approach utilizes an XLM-R (Conneau et al. 2019) transformer architecture, pre-trained on 6 M unlabeled tweets related to the sexism detection task. Additionally, we use semi-supervised learning to create an additional pseudo-labeled dataset by using unlabeled data. Finally, we inject semantic-enriched information into the system through a Sentence-BERT (SBERT) layer (Reimers and Gurevych 2019). Our approach allows the model to learn more about the underlying structures and patterns of language without relying on a large amount of labeled data.

We hypothesize that pre-training with task-related data directly will yield more effective results. In addition, previous studies have encountered challenges in handling examples in the dataset that discuss hate, political, or other controversial topics without directly addressing sexism (Rodríguez-Sánchez, Carrillo-de Albornoz, and Plaza 2020), leading to errors in their systems. To address this issue, our research is the first to incorporate semantic or topic-related information in sexism detection, enhancing the model's ability to differentiate between sexism and other types of content. To the best of our knowledge, this represents the first attempt to explore such an approach for sexism detection. Furthermore, although previous state-of-the-art works in sexism identification and categorization have utilized ensemble systems of transformers to enhance performance (de Paula, da Silva, and Schlicht 2021; Vaca-Serrano 2022); these systems depend on multiple models, making deployment and adaptation to new languages more difficult. In contrast, our approach employs a single multilingual transformer during fine-tuning (with SBERT remaining frozen), which improves efficiency and simplifies deployment.

In summary, our work aims to develop a sexism detection and categorization system that effectively addresses the challenges of limited labeled data and task-specific language variations to answer the following research questions:

1. Research Question 1: How effective are data augmentation techniques, such as pre-training and semi-supervised learning, in detecting sexism? To address this question, we apply both task-adaptation pre-training and semi-supervised learning techniques and assess their effectiveness.

2. Research Question 2: What is the impact of incorporating external multilingual semantic knowledge on model performance in sexism detection? In this regard, we employ multilingual SBERT to inject general semantic information and evaluate its influence on the system's performance.
3. Research Question 3: Does the proposed approach surpass the performance of other complex ensemble systems in the task of detecting sexism? To investigate this, we combine all the techniques and compare the results with the best-performing systems in the current state-of-the-art.

In this paper, we present several significant contributions to the field of sexism detection and categorization in online text, particularly focusing on the development of a machine learning system:

1. We systematically investigate various techniques for identifying sexism online, analyzing their efficacy and determining the most predictive approaches for use in our proposed model.
2. We design and implement a state-of-the-art model for detecting sexism in social media, incorporating novel techniques and evaluating its performance on two benchmark datasets, providing a comparison to existing approaches.
3. We conduct a deep analysis of the results to gain insight into the main challenges for this task.

The rest of this paper is organized as follows: in Section 2 we present related works in sexism detection and the application of task-adaptation and data augmentation techniques to different NLP tasks. In Section 3, the EXIST shared tasks and datasets are presented. In Section 4, the proposed classification system is presented. Section 5 presents the results, and finally, the conclusions and future works are presented in Section 6.

2 | Related Works

The detection of sexism in online conversations has become a significant and evolving area of research. While notable progress has been made in the detection of misogyny (Waseem 2016; Waseem and Hovy 2016; Frenda et al. 2019; Anzovino, Fersini, and Rosso 2018), less attention has been paid to the broader spectrum of sexism. There is often confusion between hate speech against women, misogyny, and sexism detection (Rodríguez-Sánchez, Carrillo-de Albornoz, and Plaza 2020). Sexism is a more general concept that includes attitudes, beliefs, and behaviors that perpetuate gender-based stereotypes and inequalities. It can take different forms, such as benevolent sexism, hostile sexism, and ambivalent sexism (Glick and Fiske 1996). On the other hand, misogyny is a specific form of sexism that targets women and girls and expresses hatred or contempt towards them. As a result, misogyny is a subset of sexism, and while all instances of misogyny are sexist, not all cases of sexism are misogynistic. This section reviews some of the key works related to the detection of sexism and misogyny in online conversations, as well as the main techniques employed to identify them.

2.1 | Misogyny Detection

One of the earliest attempts at detecting misogyny was made by Waseem and Hovy (2016). They proposed a supervised learning approach that classified tweets as either sexist, racist, or neither using a combination of linguistic features and traditional machine learning techniques, obtaining an F1-score of 73.89% for their best model. They produced a dataset consisting of 16,000 tweets, with only 20% of them labeled as sexist. (Waseem 2016) extended this dataset with an additional 4033 tweets and demonstrated the impact of allowing majority voting on classification and agreement between amateur and expert annotators.

Progress in online misogyny research, as a particular expression of hate speech, has been fueled by international competitions organized to identify the most effective computational methods for detecting and categorizing misogynistic content in online conversations. The Automatic Misogyny Identification (AMI) shared task (Fersini, Rosso, and Anzovino 2018) aimed to develop and evaluate computational techniques for identifying and classifying misogynistic content in online conversations. The competition comprised two tasks: classifying each text instance as either “misogynistic” or “non-misogynistic,” and if the text is “misogynistic,” classifying it into one of five subcategories: “discredit,” “harassment/threats of violence,” “derailing,” “stereotype/objectification,” and “dominance.” The approaches submitted to the competition primarily relied on supervised machine learning, focusing on various textual features (like unigrams, bigrams, sentiment-based information, or syntactic categories) or user-based features (such as the number of retweets, followers, etc.) (Pamungkas et al. 2018; Canós 2018; Nina-Alcocer 2018). The best-performing team in English (Pamungkas et al. 2018) achieved an accuracy of 0.91, and the best team for Spanish achieved an accuracy of 0.81 (Canós 2018). The SemEval 2019 (Task 5) (Basile et al. 2019) aimed to detect hate speech against immigrants and women in Spanish and English tweets. This competition used a corpus of 19,600 tweets, with 13,000 in English and 6600 in Spanish, annotated by crowdsourcing workers. The best results in all proposed tasks were achieved by traditional machine learning methods combining a variety of standard lexical and syntactic features with specific features for capturing offensive language exploiting external lexicons (Bauwelinck et al. 2019). Some participants also tested deep learning methods without much success (Paetzold, Zampieri, and Malmasi 2019).

The detection of misogyny has benefited from the development of new taxonomies and datasets. For instance, Guest et al. (2021) developed an expert-labeled dataset of 6567 messages sourced from Reddit posts and comments. They also proposed a new taxonomy for labeling misogyny that was inspired by Anzovino, Fersini, and Rosso (2018), which included four categories: misogynistic pejoratives, descriptions of misogynistic treatment, acts of misogynistic derogation, and gendered personal attacks against women. Similarly, Petrak and Krenn (2022) focused on detecting misogyny in comments from a large Austrian German language newspaper forum. They created a corpus of 6600 comments annotated with five levels of misogyny, involving forum moderators as experts in creating the annotation guidelines and annotating the comments. Additionally, they used German BERT models to evaluate their dataset, obtaining a maximum accuracy of 0.78. More recently, Richter et al. (2023) developed a new dataset consisting of 10,000 instances from English movie subtitles, categorized into 12 different classes and labeled by trained experts. New datasets for other languages have also been developed recently. For example, Glasebach et al. (2024) created a new corpus for German, and Álvarez-Crespo and Castro (2024) developed a new corpus for Galician, a less-resourced language.

2.2 | Sexism Detection

Some research has recognized the distinction between sexism and misogyny in social media. For instance, Rodríguez-Sánchez, Carrillo-de Albornoz, and Plaza (2020) created the MeTwo dataset, which is the first Spanish corpus of sexist expressions on Twitter (recently renamed to "X"). The dataset includes different forms of sexism and was labeled based on the majority vote of three annotators. The dataset comprises 3600 tweets, and the main goal is to identify instances of sexism. The authors also compared traditional and neural network-based approaches and found that BERT outperformed other tested algorithms.

Moreover, Jiang et al. (2021) introduced the first Chinese dataset for online sexism detection, the SWSR dataset, which focuses on the widely-used microblogging platform in China, Sina Weibo. Their hierarchical categorization first classifies messages as "sexist" or "non-sexist." If a text is labeled sexist, it is further categorized into "Stereotype based on Appearance", "Stereotype based on Cultural Background," "Microaggression," or "Sexual Offense." Lastly, sexist comments are classified as "Individual" (addressing a specific person) or "Generic" in the third level. They also developed a new sexism-related offensive lexicon called the SexHateLex. Similarly, Hoefels, Çöltekin, and Mădroane (2022) developed CoRoSeOf, a large corpus of Romanian social media messages annotated according to a two-level labeling scheme. The first level categorized content as either "sexist" or "non-sexist," and the second level used three categories for sexist messages ("sexist direct," "sexist descriptive," and "sexist reporting") and two categories for non-sexist posts ("non-offensive" and "offensive" posts). The final dataset includes 39,245 tweets annotated by 10 annotators (seven women and three men).

Almanea and Poesio (2022) presented the Arabic Misogyny and Sexism Corpus (ArMIS), which includes annotations from annotators with varying degrees of religious beliefs. The authors found that such differences led to disagreements in the annotation, and their work represents the first dataset to explore the effect of beliefs on misogyny and sexism annotation in detail. Recently, Altn and Saggion (2024) presented a novel dataset of approximately 7000 texts for studying sexism in Turkish. They used the same labeling schema as EXIST, and the dataset was annotated by experts. Similarly, Krenn et al. (2024) presented a sexism/misogyny dataset extracted from comments of a large online forum of an Austrian newspaper. They gathered around 8000 texts which were annotated with five levels of sexism/misogyny, ranging from 0 (not sexist/misogynist) to 4 (highly sexist/misogynist). The professional forum moderators of the online newspaper were involved as experts in the creation of the annotation guidelines and the annotation of the user comments.

In 2021 and 2022, the sEXism Identification in Social neTworks (EXIST) shared tasks were proposed at the IberLEF forum (Rodrguez-Sánchez et al. 2021, 2022). The EXIST challenges were the first to address the identification and categorization of sexism in the broadest sense, from overtly hostile to other subtle and even benevolent expressions. Previous works on sexism in online media had focused on detecting misogyny or hatred toward women (Fersini, Rosso, and Anzovino 2018; Pamungkas, Basile, and Patti 2020; Guest et al. 2021). The competition provided a benchmark dataset for the community to evaluate and compare different approaches for identifying and categorizing sexism. The whole EXIST dataset is composed of more than 12,000 tweets and gabs (messages from Gab.com) in two languages, English and Spanish. In the EXIST 2021 competition, the top-performing system was AI-UPV (de Paula, da Silva, and Schlicht 2021), which employed an ensemble of distinct transformer models, including BERT for English, BETO for Spanish, and mBERT for multilingual tasks. Furthermore, they used back-translation using the training data to augment the data available. In EXIST 2022, the best system was proposed by Avacaondata (Vaca-Serrano 2022), where their most effective approach to the task was based on an ensemble of various transformer models, including BERTweet-large, RoBERTa, and DeBERTa v3 for English, as well as BETO, BERTIN, MarIA-base, and RoBERTuito for Spanish. In both competitions, the top-performing solutions consist of large ensembles of multiple transformer models, presenting a challenging baseline due to the number of models employed and their substantial predictive capacity. However, these approaches heavily rely on large labeled datasets, involve monolingual or specific-language transformers that are not suitable when new languages need to be added, and the scale and number of transformer models make them less efficient and more difficult to implement and deploy in real-world scenarios. To address these challenges, we designed a novel architecture using a single multilingual transformer during fine-tuning (with SBERT remaining frozen) that can handle multiple languages simultaneously without modifications, improves efficiency, and simplifies deployment.

2.3 | Unsupervised Task-Adaptation and Semi-Supervised Learning for NLP

Unsupervised domain and task adaptation has shown significant advantages in various natural language processing (NLP) tasks (Gururangan et al. 2020). This technique involves adapting a pre-trained language model to a new domain or task without using labeled data, by fine-tuning the pre-trained model on data from the new domain or task, and can result in substantial improvements in performance. Unsupervised domain and task adaptation can be useful in scenarios where labeled data is scarce or expensive to obtain. For instance, Caselli et al. (2021) released HateBERT, a BERT model for abusive language detection in English re-trained on RAL-E, a large-scale dataset of Reddit comments in English from communities banned for being offensive, abusive, or hateful. They evaluated the system on different datasets and showed how it consistently outperforms a generic

BERT across different abusive language phenomena. In addition, Singh and Li (2021) explored the use of auxiliary data to improve offensive language detection using bidirectional transformers. They investigated the effectiveness of incorporating various auxiliary datasets during the pre-training phase, as well as the impact of dataset size and quality. Their findings showed that combining auxiliary data with unsupervised domain adaptation can significantly enhance the performance of offensive language detection models.

Despite recent advancements, domain adaptation remains a significant challenge in sexism detection. In an effort to address this challenge, Bose, Illina, and Fohr (2021) investigated the effectiveness of several unsupervised domain adaptation approaches for cross-corpora abusive language detection. They used Masked Language Model (MLM) pre-training to adapt different transformer models and found that the Unsupervised Domain Adaptation approaches result in suboptimal performance, while the MLM fine-tuning approach performs better in the cross-corpora setting. Similarly, Mina et al. (2021) attempted to use pre-training adaptation in the EXIST 2021 competition, but their best model, based on XLM-R with unsupervised pre-training on the EXIST data and additional hate speech-related datasets, was far from producing the best results. A potential limitation of this study may arise from using a hate speech dataset for pre-training, as sexism can also encompass non-hateful behaviors. This discrepancy could affect the model's ability to accurately capture the full spectrum of sexist expressions present in online conversations. To address this gap, we use data directly related to the same task, rather than data from a related task, such as hate speech.

Semi-supervised learning has been explored in several previous works as a way to improve the training process by augmenting the data available. One common approach is self-training, which involves training a model on a small labeled dataset and using it to predict labels for the unlabeled data. The predicted labels are then added to the training set, and the process is repeated iteratively. For instance, Alsafari and Sadaoui (2021) used self-training to improve Offensive and Hate Speech detection, where their classifier was self-trained iteratively using the most confidently predicted labels obtained from an unlabeled Twitter corpus of 5 million tweets. Abburi et al. (2021) evaluated this approach for a multi-label classification of accounts (reports) of sexism. One of the main limitations of this approach is the computational complexity associated with working with big datasets, such as the one used in our study. To mitigate this issue, we employ a technique that involves creating pseudo-labels for the unlabeled data. This is done by first training the model on the labeled dataset, and then using the trained model to generate pseudo-labels for the unlabeled data. To our knowledge, this is the first time this technique has been used for sexism detection.

This paper proposes a novel approach to improve online sexism detection and categorization by combining unsupervised task adaptation with semi-supervised learning through data augmentation. Previous successful approaches used ensembles of multiple transformers and specific models for English and Spanish, making them inefficient for real-world deployment due to the number of models and their specificity to only two languages. In contrast, our approach uses a single multilingual transformer during fine-tuning (with SBERT remaining frozen). Additionally, while previous approaches augmented data by pre-training the model on hate speech-related data, we use data specifically from the sexism domain. We hypothesize that this combination will enhance model performance by providing more labeled data and more effective adaptation to the target task. Prior research has primarily focused on hate speech detection and explored these techniques individually, but to our knowledge, no previous work has combined these techniques for online sexism detection. Furthermore, this study introduces the use of SBERT to encode the semantic information of textual content, potentially improving the model's ability to recognize subtle instances of sexism. Therefore, our proposed method overcomes the limitations of previous approaches and achieves state-of-the-art performance in detecting and categorizing sexism.

3 | Data and Resources: The EXIST 2021 and 2022 Competitions

In this section, we present a detailed overview of the sEXism Identification in Social neTworks (EXIST) challenges, including the task definitions, the evaluation procedures, and the datasets provided to participants. We also describe our methodology for collecting approximately 6 million unlabeled tweets in both English and Spanish, which we used to improve the performance of our proposed model.

3.1 | Task Definition and Evaluation

The EXIST 2021 and 2022 shared tasks are multilingual classification tasks aimed at detecting and categorizing sexism in social media posts, specifically tweets and gabs (i.e., posts from the Gab.com social network).

The challenge includes two subtasks. The first subtask (task 1) requires binary classification to determine whether a given text is sexist (i.e., it is sexist itself, describes a sexist situation, or criticizes a sexist behavior). The second subtask (task 2) aims to categorize the text according to the type of sexism (based on the expert-proposed categorization that considers different facets of women being undermined). Thus, this task is defined as a multi-class classification problem where each sexist tweet or gab must be categorized in one of the five following classes:

- **Ideological and inequality:** The text discredits the feminist movement, rejects inequality between men and women or presents men as victims of gender-based oppression.
- **Stereotyping and dominance:** The text expresses false ideas about women, suggesting they are more suitable for certain roles, inappropriate for specific tasks, or claims men's superiority over women.
- **Objectification:** The text presents women as objects apart from their dignity and personal aspects or assumes certain physical qualities women must have to fulfill traditional gender roles.

- **Sexual violence:** The text contains sexual suggestions, requests for sexual favors, or harassment of a sexual nature (rape or sexual assault).
- **Misogyny and non-sexual violence:** The text expresses hatred and violence towards women.

Task 1 was evaluated in terms of accuracy as the distribution between sexist and non-sexist categories was balanced, while task 2 was assessed using macro-averaged F1-score considering the non-sexist class. More detailed definitions along with examples for each of the categories, as well as the details of the evaluation procedure, may be found in the overviews of the EXIST challenges (Rodrguez-Sánchez et al. 2021, 2022).

3.2 | The EXIST Datasets

The EXIST 2021 and 2022 shared tasks employ data from Twitter and Gab in English and Spanish. Specifically, EXIST 2022 employs the EXIST 2021 dataset for training purposes, while a new test set labeled by experts in the task is used for the test set. Consequently, Twitter data is incorporated in both the training and testing phases, whereas Gab data is exclusively included in the EXIST 2022 training set or the EXIST 2021 test set. In this work, we separately assess the performance of the proposed system using both benchmarks to ensure a comprehensive evaluation of its effectiveness.

To gather training and testing data for the EXIST challenges, experts compiled and analyzed terms commonly used to undermine women's roles in society. The final list comprised 116 seed terms for Spanish and 109 for English, which were used to search for tweets on Twitter API. For EXIST 2021, 545,717 tweets in Spanish and 662,895 in English were collected between December 1st, 2020, and February 28th, 2021. For EXIST 2022, 170,210 tweets in Spanish and 206,549 in English were collected between January 1st and January 31st, 2022. Gabs for EXIST 2021 were extracted from Pushshift[1] (Baumgartner et al. 2020) and sampled to prepare data for labeling (Rodrguez-Sánchez et al. 2021, 2022). In this work, we followed the same procedure and extended the crawling period from November 25th, 2020, to March 10th, 2022, collecting approximately 6 million tweets in both English and Spanish, which were subsequently utilized in the pre-training phase of our proposed model.

A major difference between EXIST 2022 and its first edition in 2021 is the annotation procedure. In 2022, the test set was labeled by six trained experts, which significantly improved the quality of annotations. In contrast, the 2021 edition had a larger test set annotated through crowdsourcing using the Amazon Mechanical Turk[2] platform. Additionally, EXIST 2022 acknowledges potential differences in perception of sexism between genders and has an annotation team consisting of three women and three men to ensure a balanced representation.

The final dataset provided to participants for EXIST 2021 consists of 6977 tweets for training and a test set containing 3386 tweets and 982 gabs for testing. For EXIST 2022, the entire EXIST dataset contains 12,403 labeled texts, 11,345 for training corresponding to EXIST 2021 and a new test set consisting of 1058 tweets.

Table 1 summarizes the description of the EXIST 2022 dataset, which includes the whole EXIST 2021 dataset and the number of texts per class for both training and test sets and the distribution by language. It is important to note that we employ both EXIST 2021 and 2022 datasets separately in our experiments. Thus, for EXIST 2021 experiments, we utilize the EXIST 2021 training and test sets, while for EXIST 2022 experiments, we use the complete EXIST 2021 dataset as the training set and the EXIST 2022 test set.

4 | An Unsupervised Task-Adaptation and Semi-Supervised Data Augmentation Approach for Sexism Detection and Categorization

In this section, we present a novel cascade approach for identifying and categorizing sexism. Our approach involves a sequential process where each subsequent task takes input from the previous task. Task 1 involves binary classification of input text to determine if it contains sexism. After classification as either sexist or non-sexist, task 2 determines the type of sexism present.

To accomplish this process, our system integrates three concepts into a cohesive two-step method. First, we pre-train a multilingual transformer architecture using unlabeled tweets. Second, we fine-tune this architecture by adding a semantic layer (SBERT) and augmenting the dataset through semi-supervised learning. The overall architecture is illustrated

TABLE 1 | Distribution of instances per class and language in the training and test sets of the EXIST 2021 and 2022 datasets.

	Training (EXIST 2021 dataset)		Testing		
	Spanish	English	Spanish	English	Total
Sexist	2864	2794	254	215	6127
Non-sexist	2837	2850	271	305	6263
Ideological-inequality	768	719	97	64	1648
Misogyny-non-sexual-violence	658	499	32	25	1214
Objectification	418	406	18	21	863
Sexual-violence	375	542	44	43	1004
Stereotyping-dominance	645	628	60	55	1388

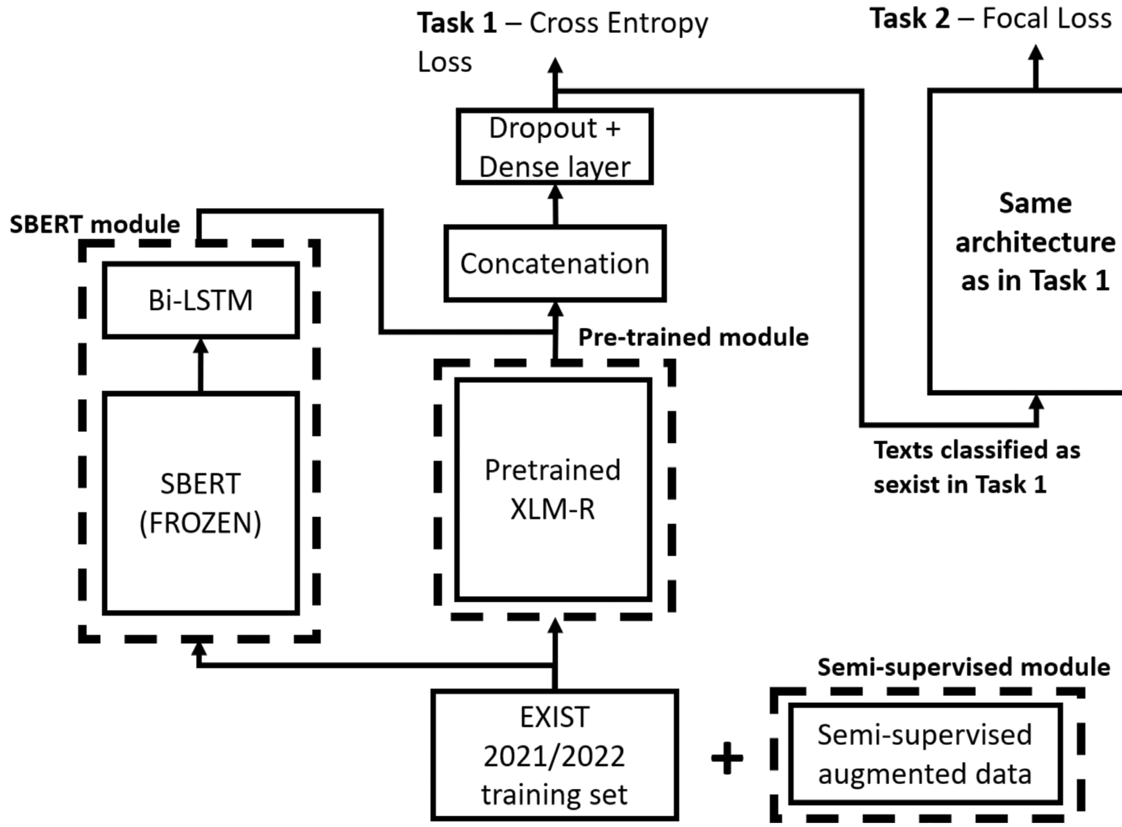

FIGURE 1 | Architecture of the proposed classification system for EXIST tasks 1 and 2, integrating SBERT, pre-trained XLM-R, and semi-supervised modules. Each module is indicated by a dotted square.

in Figure 1. As shown, our proposed classification system for EXIST tasks 1 and 2 integrates SBERT, pre-trained XLM-R, and semi-supervised modules (a dotted square indicates each module). The outputs of the SBERT and pre-trained modules are combined, while the semi-supervised module adds augmented data to the training process. Task 2 uses the same architecture but is trained exclusively on texts classified as sexist. During inference, the input to the Task 2 module consists only of the texts identified as sexist in Task 1.

4.1 | Unsupervised Task Adaptation

In the first step, we use unsupervised task-adaptation techniques (TAPT) to transfer knowledge from a large-scale task-unlabeled dataset to the target task without requiring additional labeled target data. Unlike domain-adaptive pre-training (DAPT), which typically uses a larger unlabeled dataset from a related domain (e.g., offensive language or hate speech), TAPT strikes a different balance. It utilizes a smaller pre-training corpus that is more task-relevant, making it substantially less resource-intensive than DAPT. Its performance is frequently on par with that of DAPT (Gururangan et al. 2020). We hypothesize that using a smaller, task-relevant pre-training corpus better equips the model to capture the nuances and specific characteristics of sexism detection, rather than relying on a large-scale, unrelated corpus.

We employ a masked language modeling (MLM) objective to leverage the benefits of unsupervised task adaptation. In this approach, we pre-train an XLM-R transformer model using cross-entropy loss on predicting random tokens. Specifically, 15% of the tokens in the input text are randomly masked, and the model is trained to predict the masked tokens based on the context provided by the remaining unmasked tokens. This process helps the model in learning contextualized representations of words in different contexts. It is important to mention that we utilize the whole word masking technique (Cui et al. 2021), which consists of masking all subwords corresponding to a word at once to ensure that the model captures linguistic patterns at the word level rather than subword units. For this process, we utilized the unlabeled Twitter data described in Section 3 during the pre-training phase. We implemented this method using two NVIDIA RTX 3090 GPUs. To optimize the model, we used the Adam (Kingma and Ba 2015) optimizer with an initial learning rate of 1e-4 and a linear weight decay of 0.01. We trained the model using a batch size of 32 for 10 epochs. Figure 2 shows the training loss curve for this model.

As shown in Figure 1, the pre-trained module is applied directly to the input text in the fine-tuning process. Before training, we preprocess the data by replacing all URLs and usernames with “*twurl*” and “*twuser*,” respectively. The pre-trained module XLM-R captures contextual information in the text. Given an input text x, XLM-R generates a contextualized representation h_{xlmr}:

$$h_{xlmr} = XLMR(x; \Theta_{xlmr})$$

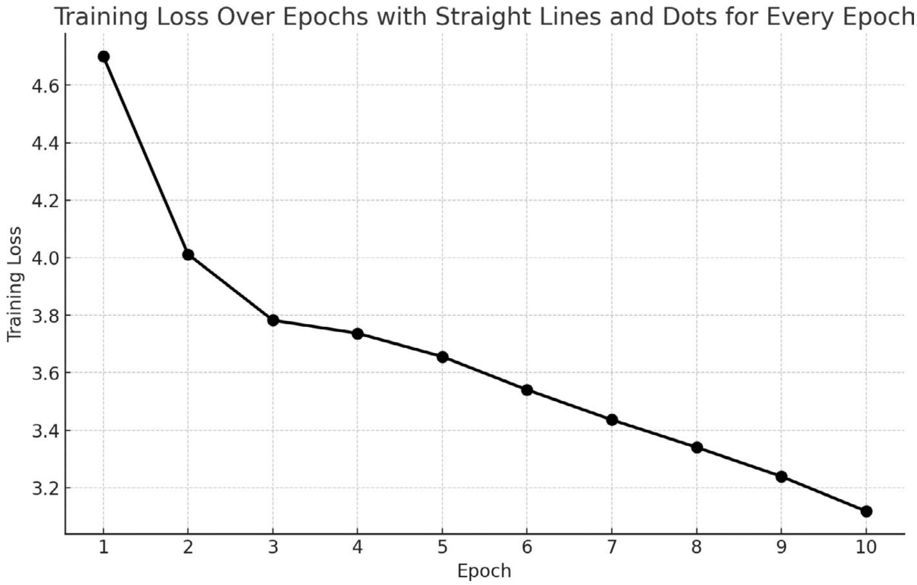

FIGURE 2 | Training loss curve over 10 epochs for the unsupervised task-adaptation process (TAPT).

where $XLMR(\cdot)$ is the pre-trained transformer base XLM-R, and Θ_{xlmr} denotes the model parameters pretrained and adapted during fine-tuning.

4.2 | Data Augmentation via Semi-Supervised Learning

Semi-supervised learning is a technique that can train models on datasets with only a small proportion of labeled examples, where the majority of the data is unlabeled. In the context of detecting sexism online, this method is particularly useful, as obtaining large labeled datasets for this task can be challenging. To augment the labeled data available for sexism detection, we employ a pseudo-labelling semi-supervised learning approach. This approach uses both labeled and unlabeled data to generate new pseudo-labeled instances. We use the original pre-trained XLM-R transformer model to create two additional datasets of 20,000 tweets each for EXIST 2021 and EXIST 2022 collections. We generate these datasets by applying the pre-trained model to the unlabeled tweets described in Section 3, which assigns pseudo-labels to the unlabeled data based on the model's understanding of both tasks.

One main issue of this process is the potential propagation of incorrect labels. Since the model generates labels for the unlabeled data, any mistakes or biases present in the initial model may be amplified, leading to a decreased overall performance or increased bias in the final model. To mitigate this issue, we set high confidence thresholds of 0.99 for task 1 and 0.6 for task 2 when selecting tweets for inclusion in our training datasets. This ensured that the selected tweets were likely to be accurately labeled, contributing to the reliability of the semi-supervised learning process. After filtering the tweets based on the confidence level, we sampled these high-confidence predictions to create two datasets for EXIST 2021 and 2022, each containing 20,000 tweets. This process was done separately for both EXIST 2021 and EXIST 2022 to ensure we do not employ information from the test set in EXIST 2021.

By incorporating these additional datasets into our training process, we aim to enhance the model's performance and generalization by exposing it to a more diverse range of examples, which can help it better understand and classify sexism-related content. Unlike self-training, which repeatedly fine-tunes the model on its own generated labels, our approach combines the outputs of the original model to achieve a more robust and accurate representation of the underlying patterns in the data. This can help to reduce the risk of introducing noisy or erroneous labels into the training process, while also saving computational resources. By selecting only high-confidence predictions to create the new datasets, we further ensure the quality and reliability of the augmented data.

4.3 | Semantic-Enriched Information Layer

To capture the semantic nuances of sexist content more effectively, we incorporate additional semantic-enriched information into our feature representation. Sentence-BERT (SBERT) is a powerful pre-trained model for computing sentence embeddings that capture the semantic meaning of sentences (Reimers and Gurevych 2019). These embeddings capture the semantic relationships between words and the overall meaning of sentences more effectively. As a result, this layer can help our system better identify and understand the linguistic nuances of sexist content, leading to improved detection and categorization.

In our approach, we utilize a layer of SBERT, which is frozen during the fine-tuning process to preserve its pre-trained semantic understanding. In particular, we used a multilingual version based on XLM-R.[3] This layer is followed by an unfrozen LSTM layer, which helps capture the contextual information and dependencies within the input text at the training step.

Like the pre-trained module, the SBERT module is applied to the input text after preprocessing. The SBERT module is specifically designed to produce sentence-level embeddings. Thus, given the exact input text x as in the pre-trained module, SBERT generates a contextualized representation h_{sbert} that will be fixed during fine-tuning for the same input:

$$h_{sbert} = SBERT(x; \Theta_{sbert})$$

where $SBERT(\cdot)$ is the pre-trained transformer base SBERT, and Θ_{sbert} denotes the pre-trained model parameters that will be fixed during fine-tuning. The SBERT module adds a Bi-LSTM layer to the output (Hochreiter and Schmidhuber 1997), which enhances the system's ability to capture long-range dependencies and sequential information in the text:

$$h_{bilstm} = BiLSTM(h_{sbert}; W, U)$$

where W and U are learnable weight matrices of the network that will be trained during fine-tuning.

4.4 | Classification

After the input x has been propagated through the pre-trained and SBERT module, the output is concatenated:

$$h_{concatenated} = [h_{bilstm}, h_{xlmr}]$$

This combined representation enables our system to benefit from the powerful contextual understanding of the XLM-R model in the sexist task context and the semantic-rich representations provided by the SBERT and Bi-LSTM layers. Finally, we add a dropout layer, a fully connected layer and apply the softmax function to serve as the output probabilities of each class:

$$h_{dropout} = dropout(h_{concatenated}, dropout\_rate)$$
$$logits = dense(h_{dropout}; W)$$

$$\hat{y}_i = \frac{e^{logits_i}}{\sum_{j=1}^{K} e^{logits_j}}$$

where $dropout\_rate$ is set to 0.1, W is a learnable weight matrix that will be trained in the fine-tuning process and K is equal to the number of classes.

Cross-entropy is used as the final loss function for task 1:

$$L(y, \hat{y}) = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

where C is the number of classes (2 for task 1) and y_i is the ground truth label.

In task 2, the standard cross-entropy loss may not be sufficient to handle imbalances between the classes (see Table 1), as the model tends to prioritize learning the majority class, leading to poor performance in minority classes. Therefore, we decided to employ Focal loss (Lin et al. 2017). The focal loss is an extension of the cross-entropy loss, that introduces a modulating factor to ensure that the model focuses more on hard-to-classify examples while reducing the contribution of easy examples to the total loss. This modulating factor is controlled by a tunable focusing parameter, γ. The focal loss can be expressed as:

$$L(y, \hat{y}) = -\sum_{i=1}^{C} y_i (1 - \hat{y}_i)^{\gamma} \log(\hat{y}_i) \quad (1)$$

where C is the number of classes (5 for task 2) and y_i is the ground truth label. In all our experiments, the parameter γ is set to 2.

5 | Results and Discussion

We evaluate our proposed approach for detecting sexism on two benchmark datasets: EXIST 2021 and EXIST 2022. To establish a baseline, we first fine-tune a base XLM-R model with the training data. We then assess the performance of the base model by gradually incorporating each of our proposed techniques and comparing the results. Finally, we compare our system's performance against the top-performing submissions from the EXIST 2021 and 2022 shared tasks to provide a comprehensive evaluation of our approach in comparison to existing state-of-the-art methods. To gain deeper insights into our system's performance and characteristics, we conduct three manual evaluations of the results.

5.1 | Experimental Setup

In the fine-tuning phase of our experiments, we employed an NVIDIA RTX 3090 GPU for computation. Optimization was conducted using the Adam optimizer (Kingma and Ba 2015), with an initial learning rate of 8e-6 and a linear weight decay of 0.01 for all experiments involving the XLM-R pre-trained model. For the experiments using the baseline (XLM-R without task-adaptation), we utilized an initial learning rate of 2e-5 and a linear weight decay of 0.01. In both cases, the models were trained with a batch size of 16 for three epochs. Our implementation was executed using the PyTorch library (Paszke et al. 2017) and the Transformers library provided by HuggingFace (Wolf et al. 2019). Figures 3 and 4 illustrate the training loss (black line, left y-axis) and test accuracy (blue line, right y-axis) across three epochs for our best-trained systems on both tasks for EXIST 2021 and 2022, respectively. The plots show a reduction in training loss and an increase in test accuracy over time, highlighting the effectiveness of the system's training process for both tasks.

To address the feasibility of deploying our proposed system in real-world applications where access to high-performance hardware might be limited, we conducted a benchmarking analysis to evaluate the inference time on both GPU and CPU environments. Specifically, we measured the time required for each iteration during inference using an NVIDIA RTX 3090 GPU and an Intel Core i9-10920X CPU. The benchmarking results, presented in Table 2, show that each iteration takes approximately 75.8 ms on the GPU and 102.27 ms on the CPU. Although the GPU offers faster processing times, the CPU performance remains within an acceptable range for practical applications. This indicates that our system can be effectively deployed even in environments where high-performance GPUs are not available.

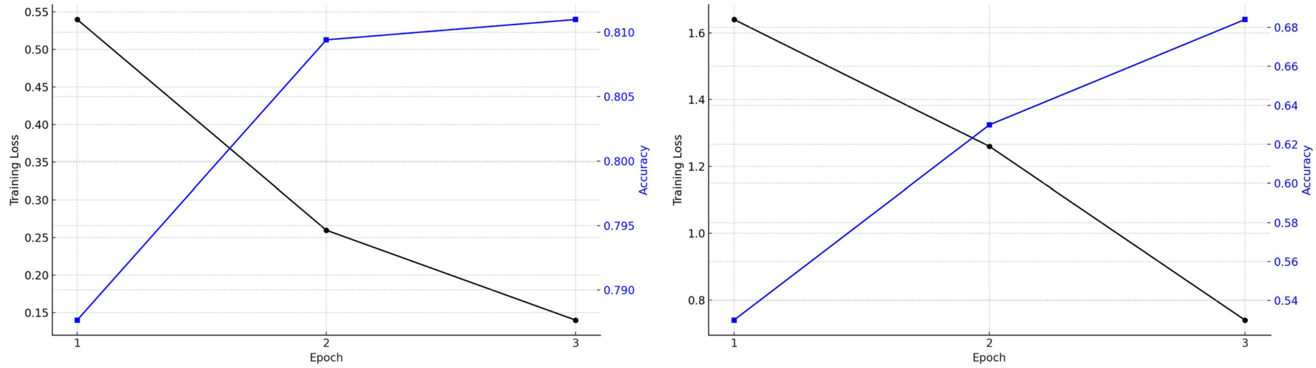

FIGURE 3 | Training loss and test accuracy per epoch for our best system in EXIST 2021 (left: Task 1, right: Task 2).

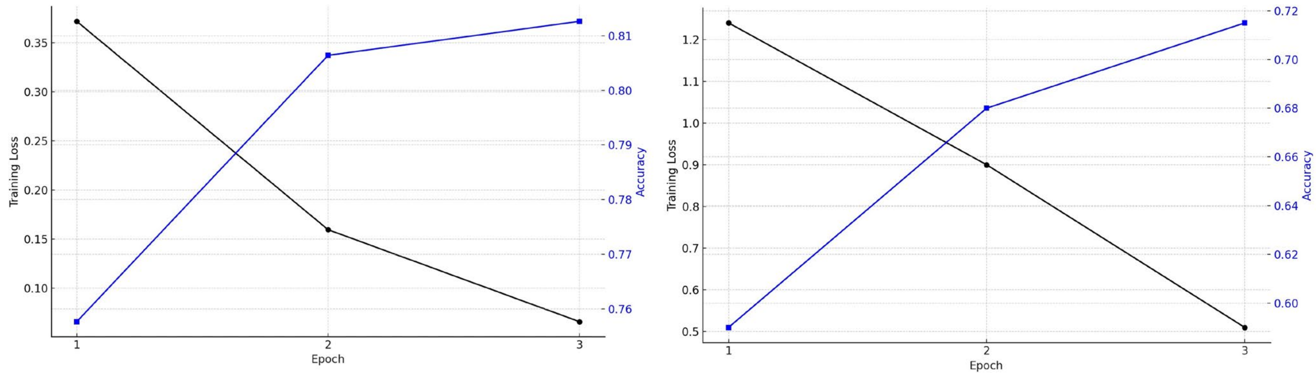

FIGURE 4 | Training loss and test accuracy per epoch for our best system in EXIST 2022 (left: Task 1, right: Task 2).

TABLE 2 | Benchmarking results for inference time.

Hardware environment	Inference time per iteration (ms)
NVIDIA RTX 3090 GPU	75.8
Intel Core i9-10920X CPU	102.27

5.2 | Evaluation Results

Tables 3 and 4 present the results of our experiments on EXIST 2021 tasks 1 and 2, respectively. The effectiveness of our approach is evaluated using four metrics: accuracy, precision, recall, and F1 score. However, to facilitate the comparison of our techniques, we report accuracy for Task 1 and F1 score for Task 2, as these were the evaluation metrics used in the competition for ranking purposes. By using the same evaluation criteria as the original competition, we ensure that our analysis is consistent and can be directly compared to the performance of other submissions.

For EXIST 2021 task 1 (see Table 3), the combination of task-adaptation (XLM-R pre-trained), data augmentation via semi-supervised learning (semi-supervised), and SBERT outperformed all other methods, achieving an accuracy of 0.812. This represents a 5.1% increase in accuracy compared to the baseline XLM-R model. Analyzing individual contributions, the addition of SBERT to the baseline model resulted in a slight 0.5% increase in accuracy. On the other hand, data-driven strategies such as task adaptation and semi-supervised learning contributed more substantially to the performance improvement, with increases of 3.5% and 2.9%, respectively.

Similarly, for EXIST 2021 task 2 (see Table 4), the best performance was once again achieved by combining XLM-R pre-trained, semi-supervised learning, and SBERT, with an F1-score of 0.615. This represents an 8% increase in F1-score compared to the baseline. As observed in task 1, SBERT's contribution to the performance improvement was less significant compared to the pre-trained XLM-R and semi-supervised learning approaches.

Additionally, Tables 5 and 6 present a comparison between the performance of our most effective system and the top-performing system from the shared task, using the competition metrics. For Task 1, our system achieved an accuracy of 0.812, a 3.2% increase compared to the best system's accuracy of 0.78. Similarly, our system demonstrates better performance in different metrics. In Task 2, our combined system also shows better performance than the top system from EXIST 2021 by 3.7%. It is important to note that the performance difference among the top-5 teams in the competition is approximately 1.3% for task 1 in terms of accuracy and 2.6% for task 2 in terms of F1-score (Rodrguez-Sánchez et al. 2021). Given this narrow margin, the improvement achieved by our proposed system is of significant importance.

The results of our experiments for EXIST 2022 tasks 1 and 2 are illustrated in Tables 7 and 8, respectively. Moreover, Tables 9 and 10 present a comparison between the performance of our most effective system and the top-performing system from the shared task.

TABLE 3 | Results of EXIST 2021 task 1, baseline and comparison of techniques.

Technique	Accuracy	F1	Recall	Precision
XLM-R (baseline)	0.761	0.760	0.761	0.760
Baseline + SBERT	0.766 (↑ 0.5%)	0.766	0.767	0.766
Baseline + semi-supervised	0.790 (↑ 2.9%)	0.790	0.790	0.790
Baseline + SBERT + semi-supervised	0.793 (↑ 3.2%)	0.792	0.791	0.796
XLM-R pre-trained	0.796 (↑ 3.5%)	0.795	0.795	0.798
XLM-R pre-trained + SBERT	0.803 (↑ 4.2%)	0.801	0.800	0.804
XLM-R pre-trained + semi-supervised	0.806 (↑ 4.5%)	0.805	0.804	0.806
XLM-R pre-trained + semi-supervised + SBERT	**0.812 (↑ 5.1%)**	**0.811**	**0.810**	**0.813**

Note: The bold values represent the best result.

TABLE 4 | Results of EXIST 2021 task 2, baseline and comparison of techniques.

Technique	F1	Accuracy	Recall	Precision
XLM-R (baseline)	0.535	0.620	0.534	0.539
Baseline + SBERT	0.556 (↑ 2.1%)	0.638	0.550	0.563
Baseline + semi-supervised	0.593 (↑ 5.8%)	0.666	0.599	0.589
Baseline + SBERT + semi-supervised	0.590 (↑ 5.5%)	0.657	0.600	0.579
XLM-R pre-trained	0.600 (↑ 6.5%)	0.620	0.600	0.600
XLM-R pre-trained + SBERT	0.604 (↑ 6.9%)	0.670	0.613	0.599
XLM-R pre-trained + semi-supervised	0.609 (↑ 7.4%)	0.678	0.615	0.605
XLM-R pre-trained + semi-supervised + SBERT	**0.615 (↑ 8%)**	**0.684**	**0.625**	**0.610**

Note: The bold values represent the best result.

TABLE 5 | Results of EXIST 2021 task 1, comparison of our best system and best system of the shared task.

Technique	Accuracy	F1	Recall	Precision
Best system of the shared task EXIST 2021	0.780	0.780	0.780	0.780
XLM-R pre-trained + semi-supervised + SBERT	**0.812 (↑ 3.2%)**	**0.811**	**0.810**	**0.813**

Note: The bold values represent the best result.

TABLE 6 | Results of EXIST 2021 task 2, comparison of our best system and best system of the shared task.

Technique	F1	Accuracy	Recall	Precision
Best system of the shared task EXIST 2021	0.578	0.657	0.577	0.581
XLM-R pre-trained + semi-supervised + SBERT	**0.615 (↑ 3.7%)**	**0.684**	**0.625**	**0.610**

Note: The bold values represent the best result.

Similar to the results obtained for the EXIST 2021 task 1, we observed that all methods for EXIST 2022 task 1 surpassed the baseline performance. The combination of XLM-R pre-trained model, semi-supervised learning, and SBERT achieved the highest performance, with an accuracy of 0.811, representing a 4.7% increase compared to the baseline XLM-R model. Individually, when added to the baseline model, SBERT contributed to a 0.6% increase in accuracy. The integration of semi-supervised learning into the baseline model led to a 3% improvement, while the pre-trained XLM-R model exceeded the baseline performance with a 3.9% increase in accuracy.

For EXIST 2022 task 2, we observe similar patterns again. The combination of XLM-R pre-trained, semi-supervised learning, and SBERT achieved the best performance with an F1-score of 0.522, representing a 5.2% increase compared to the baseline. Analyzing individual contributions, the addition of SBERT to the baseline model resulted in a slight 1.7% increase in F1-score.

TABLE 7 | Results of EXIST 2022 task 1, baseline and comparison of techniques.

Technique	Accuracy	F1	Recall	Precision
XLM-R (baseline)	0.764	0.764	0.767	0.765
Baseline + SBERT	0.770 (↑ 0.6%)	0.771	0.772	0.770
Baseline + semi-supervised	0.794 (↑ 3%)	0.793	0.793	0.792
Baseline + SBERT + semi-supervised	0.791 (↑ 2.7%)	0.790	0.791	0.789
XLM-R pre-trained	0.803 (↑ 3.9%)	0.802	0.804	0.802
XLM-R pre-trained + SBERT	0.806 (↑ 4.2%)	0.805	0.806	0.804
XLM-R pre-trained + semi-supervised	0.807 (↑ 4.3%)	0.806	0.806	0.806
XLM-R pre-trained + semi-supervised + SBERT	**0.811** (↑ **4.7%**)	**0.810**	**0.811**	**0.809**

Note: The bold values represent the best result.

TABLE 8 | Results of EXIST 2022 task 2, baseline and comparison of techniques.

Technique	F1	Accuracy	Recall	Precision
XLM-R (baseline)	0.470	0.660	0.500	0.450
Baseline + SBERT	0.472 (↑ 0.2%)	0.675	0.500	0.460
Baseline + semi-supervised	0.495 (↑ 2.5%)	0.695	0.520	0.488
Baseline + SBERT + semi-supervised	0.507 (↑ 3.7%)	0.697	0.535	0.496
XLM-R pre-trained	0.510 (↑ 4%)	0.702	0.540	0.480
XLM-R pre-trained + SBERT	0.514 (↑ 4.4%)	0.709	0.540	0.500
XLM-R pre-trained + semi-supervised	0.520 (↑ 5%)	0.712	0.548	0.500
XLM-R pre-trained + semi-supervised + SBERT	**0.522** (↑ **5.2%**)	**0.715**	**0.550**	**0.500**

Note: The bold values represent the best result.

TABLE 9 | Results of EXIST 2022 task 1, comparison of our best system and best system of the shared task.

Technique	Accuracy	F1	Recall	Precision
Best system of the shared task EXIST 2022	0.799	0.797	0.797	0.798
XLM-R pre-trained + semi-supervised + SBERT	**0.811** (↑ **1.2%**)	**0.810**	**0.811**	**0.809**

Note: The bold values represent the best result.

TABLE 10 | Results of EXIST 2022 task 2, comparison of our best system and best system of the shared task.

Technique	F1	Accuracy	Recall	Precision
Best system of the shared task EXIST 2022	0.510	0.700	0.520	0.500
XLM-R pre-trained + semi-supervised + SBERT	**0.522** (↑ **1.2%**)	**0.715**	**0.549**	**0.510**

Note: The bold values represent the best result.

The addition of semi-supervised learning to the baseline model resulted in a 3.6% increase in F1-score and the pre-trained XLM-R model outperformed the baseline with a 4.2% increase in F1-score.

Comparing our best system and the best system of the EXIST 2022 competition for task 1 (sexism detection), we observe that our proposed system demonstrates superior performance in all evaluation metrics. Our system achieved an accuracy of 0.811, a 1.2% increase compared to the best system's accuracy of 0.799. In task 2 (sexism categorization), our combined system again outperforms the best system from EXIST 2022 by 1.2%. We observe improvements in all metrics again. It is important to note that the performance difference between the top-5 teams is approximately 1.1% for task 1 in terms of accuracy and 1.4% for task 2 in terms of F1-score. Given this narrow margin, the improvement achieved by our proposed system holds significant relevance.

TABLE 11 | Average improvement of different techniques over the baseline, showing the effects of combining SBERT, semi-supervised learning, and XLM-R pre-trained models.

Technique	Avg. Imp.
Baseline + SBERT	0.85%
Baseline + semi-supervised	3.55%
Baseline + SBERT + semi-supervised	3775%
XLM-R pre-trained	4475%
XLM-R pre-trained + SBERT	4925%
XLM-R pre-trained + semi-supervised	5.3%
XLM-R pre-trained + semi-supervised + SBERT	5.75%

The results demonstrate that the combination of SBERT, semi-supervised learning, and task-adaptation consistently outperforms all other approaches and the baseline model in both the EXIST 2021 and EXIST 2022 datasets, resulting in substantial improvements in both accuracy and F1-score across all tasks. Specifically, the individual contributions of each technique highlight the crucial roles played by task-adaptation and semi-supervised learning in enhancing system performance. Although SBERT also contributes to the enhancement, its impact is comparatively smaller than the other two methods.

5.3 | Discussion

To gain more insight into the contribution of each component to the system's performance, Table 11 provides an analysis of the individual contributions of the three techniques employed in our study, showing the average improvement per method with respect to the baseline for all tasks combined.

According to the table, SBERT contributes an average improvement of 0.85%. This result indicates that the sentence embeddings generated by SBERT have helped the system better capture the linguistic nuances of the text. The addition of semi-supervised learning to the baseline model leads to a more substantial average improvement of 3.55%, highlighting the efficacy of using augmented data to enhance the system's performance. Lastly, as a standalone method, XLM-R pre-training shows the highest average improvement of 4475% when compared to the baseline model, highlighting the importance of task-adaptation in boosting the model's performance.

Our analysis shows that task-adaptation and semi-supervised augmented data play crucial roles in driving improvements, while SBERT, though contributing less than the other two methods, still provides valuable complementary information to enhance the overall performance. These results underscore the importance of leveraging unlabeled data to augment the model's understanding of the problem domain.

The combination of all three techniques yields the highest average improvement (5.75%), highlighting the power of integrating all techniques to maximize performance. Contrasting the individual contributions, the impact of SBERT and semi-supervised learning on the pre-trained XLM-R model appears to be more similar in magnitude.

To better understand and explain the reasons behind the improvements provided by each specific component, we conducted an analysis focusing on the differences in classifications when using models that employ only one technique at a time. Table 12 presents the most representative text examples, along with their true labels and the predicted classifications made by the three different models: SBERT, semi-supervised, and XLM-R pre-trained.

The first three examples show relevant cases where SBERT accurately predicts the text. Most of the cases we analyzed, where SBERT outperformed the other techniques involve non-sexist texts that discuss controversial topics or hate. The three examples shown in the table discuss polemic subjects such as abortion, racism, or politics. In these instances, the SBERT technique helps the system correctly identify examples that are not sexist, even if they are hateful. The next three examples are cases where the semi-supervised technique worked better. For this technique, we identified in most cases that semi-supervised learning is helpful when data is scarce, especially in the minority classes. For instance, examples 4 and 5 involve objectification, the least frequent class (see Table 1) and semi-supervised technique helps correctly predicting these instances. Example 6 presents a text discussing feminism in a non-negative way, whereas in EXIST, there are many examples of texts discussing feminism that are sexist (with the class ideological-inequality being the majority). Therefore, using data augmentation techniques helps reduce this bias. Finally, in examples 7, 8, and 9, pre-training performs better. This technique shares many properties with semi-supervised learning, as it involves data augmentation, and their contributions are similar, as shown in Table 11. However, in this case, we identified many instances of sarcasm and jokes. All three examples illustrate these cases, and this technique seems to contribute more overall, even in these particularly challenging or ambiguous cases.

Therefore, the results can be attributed to the specific strengths of each method and their synergy when combined. SBERT generates more general context-aware sentence embeddings, which capture the semantic relationships between words and the overall meaning of sentences in a different context from sexism, and are particularly helpful when discussing hateful or polemic topics that might not be sexist. However, the contribution of SBERT is relatively smaller than the other two methods when used in isolation. On the other hand, the extra data augmented via semi-supervised learning helps the model generalize better and achieve higher performance by increasing the data at the training step. This approach is particularly effective when labeled data is scarce, as is the case with the minority classes. Finally, unsupervised task-adaptation exploits unlabeled data for model refinement, leading to the capture of nuances and specific characteristics of sexism detection, and works even with more challenging cases. By combining these three methods, we take advantage of the unique strengths of each approach while mitigating their respective weaknesses. The synergy created by this combination leads to the notable enhancements in accuracy and

TABLE 12 | Representative examples for each method used in the study (bolded predictions indicate the correct model classifications matching the true label).

No.	Text	Real	Prediction		
			SBERT	Semi-supervised	XLM-R pre-trained
1	No woman can call herself free who does not control her own body. #AbortionStandJA	Non-sexist	**Non-sexist**	Stereotyping-dominance	Stereotyping-dominance
2	@twuser @twuser according to them, it empowers you, and if you wear a burka, you are the freest woman in the world	Non-sexist	**Non-sexist**	Misogyny-non-sexual-violence	Stereotyping-dominance
3	@twuser It's the left wing middle class. They hate working class socialists like me and they hate the right wing working class But they love black lives matter feminism, the metoo movement	Non-sexist	**Non-sexist**	Ideological-inequality	Ideological-inequality
4	@twuser you look like a prostitute	Objectification	Misogyny-non-sexual-violence	**Objectification**	Misogyny-non-sexual-violence
5	@twuser then eat something else, fatty	Objectification	Non-sexist	**Objectification**	Misogyny-non-sexual-violence
6	@twuser I apologize to any feminists who have spoken out about this issue. I have learned that it's a big issue for feminists, and I'm glad to learn that. I think more people, including celebrities who champion women	Non-sexist	Ideological-inequality	**Non-sexist**	Ideological-inequality
7	First it's a day without immigrants. Now we have a day without women. Why don't we have a day without liberals. We could really get some things done then. ~Quoted from another site	Ideological-inequality	Non-sexist	Non-sexist	**Ideological-inequality**
8	I'm straight but i'm also allergic to women lol haha	Misogyny-non-sexual-violence	Stereotyping-dominance	Non-sexist	**Misogyny-non-sexual-violence**
9	No more free drinks girls ... twurl #mgtow	Misogyny-non-sexual-violence	Non-sexist	Sexual-violence	**Misogyny-non-sexual-violence**

Note: Bold indicates the correct prediction.

TABLE 13 | Comparison of classifications between our system and the best system from EXIST 2021.

No.	Text	Real	Predicted by best EXIST 2021 system	Predicted by our system
1	And they learn that they can't trust anything women actually say. If they express appreciation for a woman, they discover that this makes them seem weak and unattractive. Do you see how the big man that you're complaining about is *men reacting to the way women treat them?* ffs @twuser	Misogyny-non-sexual-violence	Stereotyping-dominance	Misogyny-non-sexual-violence
2	@twuser No... there is no reason to, from a Males stance. Every law is stacked against men #MGTOW	Ideological-inequality	Non-sexist	Ideological-inequality
3	Man left with criminal record after calling wife a 'slapper' it's illegal to be unfriendly to your wife after she's cheated on you	Misogyny-non-sexual-violence	Sexual-violence	Misogyny-non-sexual-violence
4	Honestly... what annoys me is that when niggas wanna do more damage to women that have already been through a lot and already damaged inside smh #LetHerHeal #LeaveHerAlone	Non-sexist	Misogyny-non-sexual-violence	Non-sexist
5	Why are you so obsessed with men having sex with other men? Seriously, your response to everything involves accusations of homosexuality or cocksucking. You are clearly a repressed faggot	Non-sexist	Sexual-violence	Non-sexist

TABLE 14 | Comparison of classifications between our system and the best system from EXIST 2022.

No.	Text	Real	Predicted by best EXIST 2022 system	Predicted by our system
1	@twuser Make that ass clap	Sexual-violence	Non-sexist	Sexual-violence
2	Women will hit you up just to ignore you for 2 good days. @twuser @ twuser @twuser How does it feel to literally be the thing you hate?	Stereotyping-dominance	Misogyny-non-sexual-violence	Stereotyping-dominance
3	Look at all these little keyboard warriors that turn toxically masculine when it suits them. Crying doesn't make you any less of a man	Non-sexist	Ideological-inequality	Non-sexist
4	How interested are women hahaha	Stereotyping-dominance	Non-sexist	Stereotyping-dominance
5	I can't respect a man who take his kids around every female he messing with Relationship I understand but other than that ... hell naw .... Women too	Non-sexist	Stereotyping-dominance	Non-sexist

F1-score that we observe when compared to the performance of the baseline model and the individual techniques when used in isolation.

To gain a deeper understanding of the limitations and error types within our proposed approach, we conducted various manual error analyses. First, we compared a selection of our correct predictions with the errors made by the top-performing system in EXIST 2021. Second, we carried out a similar comparison with the best system from EXIST 2022. Lastly, we examined the errors that were transferred and retained within our proposed system. Through these investigations, we identified several factors that contribute to the challenges inherent in this task.

Table 13 displays a selection of errors committed by the top-performing model in the EXIST 2021 competition, which our proposed system managed to classify correctly. This demonstrates our model's ability to categorize a range of complex examples accurately. Likewise, Table 14 showcases a series of errors made by the best-performing model in the EXIST 2022 competition that our proposed system was able to identify correctly. The table provides evidence that our model accurately categorized various complex examples that require deeper semantic understanding of the sexist language, further emphasizing the improved performance of our proposed system over the top models from both EXIST 2021 and EXIST 2022 competitions.

Table 15 presents some examples of misclassifications made by our system, which we manually examined. In the first example, our classifier struggles to detect a challenging instance of indirect sexism, where the narrative is inverted, presenting men as victims of oppression. This complexity makes it difficult for the model to identify the situation correctly. The second and third examples represent a category frequently observed during manual inspection: numerous false negatives involve tweets containing sarcasm or irony, which can pose classification challenges for the model, even if some cases are correctly classified, as seen in examples 7, 8, and 9 in Table 12. The fourth example illustrates a case in which the tweet could exhibit hate but not sexism. The inclusion of the SBERT layer partially addressed these cases (for instance, the three first examples in 12 and the fifth example in 13); however, some persist. Lastly, we encounter a recurrent example of a tweet where the combination of its length and the usage of hateful language leads our classifier to identify it as sexist erroneously.

6 | Conclusions and Future Work

In this paper, we have introduced a novel approach for detecting and categorizing sexism on social media by leveraging unsupervised task-adaptation, semi-supervised learning, and semantically-enriched information. Our findings demonstrate that this approach effectively addresses the challenges associated with limited labeled data and task-specific language variations, leading to improved performance in binary classification and 5-classes categorization tasks.

Our first research question sought to determine whether data augmentation techniques, such as pre-training and semi-supervised learning, contribute to the detection of sexism. Our study demonstrates that these techniques are beneficial individually, and their combined implementation further enhances the effectiveness of our sexism detection system. By harnessing the benefits of unlabeled data, our approach allows the model to learn more about the underlying structures and patterns of language without relying on a large amount of labeled data. Specifically, we found how task-adaptive pre-training offers a more focused and efficient approach to model pre-training, yielding competitive performance in the specific domain of sexism detection while reducing the computational

TABLE 15 | Examples of misclassifications made by our system.

No.	Text	Real	Predicted by our system
1	Yesterday, here in MA. A 45 year old woman driving the wrong way on 495 in Westford killed a 35 year old man at 2 AM. This morning the police released the name of the man killed. Still won't release the woman's name. I'm going to bet she is here illegally. Any takers?	Ideological-inequality	Non-sexist
2	*Woman detected* *Running facial recognition algorithm* *Current drivers license for target found in database* *Calculating probability of woman driving in next 24 h* *87%* *Self Defense Protocols initiated*	Stereotyping-dominance	Non-sexist
3	Woman will be driving and forget to stop at zebra crossings but they'll flash their lights to apologize so it's okay	Stereotyping-dominance	Non-sexist
4	Call me ageist but I don't think an 83 year-old should be Olympics chief? Just retire yourself and your sexist views already	Non-sexist	Stereotyping-dominance
5	Fuck that cunt state	Non-sexist	Misogyny-non-sexual-violence

demands associated with traditional domain-adaptive pre-training techniques.

In addressing the second research question, we integrated multilingual SBERT to incorporate general semantic information into our system and evaluated its influence on the performance of our sexism detection approach. Our findings reveal that, although the individual contribution of SBERT to the overall system is not as pronounced as that of the data augmentation techniques, it can be effectively combined with these methods to improve the system's performance. The integration of SBERT allows our model to identify and understand the linguistic nuances of sexist content, further enhancing the robustness of our proposed approach in addressing the challenges of sexism detection on social media.

Lastly, by addressing the third research question, we have demonstrated that the combination of task-domain adaptation, semi-supervised learning, and SBERT consistently improves performance across both tasks and datasets. Our best systems outperform the top systems that participated in the shared tasks, showcasing the effectiveness of our proposed approach. We have defined the best system and established new state-of-the-art results for all tasks and both benchmarks, EXIST 2021 and 2022. Our system, in addition to achieving the best results, is also more efficient, as our approach optimizes a single multilingual transformer during fine-tuning (with SBERT being frozen), which enhances efficiency and facilitates deployment. Moreover, by employing a task-adaptation technique using a smaller pre-training corpus that is more relevant to the target task, the model is better equipped to capture the nuances and specific characteristics of sexism detection, as opposed to relying on a large-scale unrelated corpus. In conclusion, the combination of SBERT, semi-supervised learning, and XLM-R pre-trained models demonstrates significant effectiveness in identifying sexist language in social media text, thereby providing a promising framework for future research and practical applications in detecting and mitigating online hate speech and harassment.

Our work has several limitations. While our study addresses the challenge of limited labeled data through semi-supervised learning and unsupervised task adaptation, it is crucial to consider potential biases in the data, particularly those arising from the pseudo-labeling process. For instance, if the model is initially trained on data that over-represents certain demographic groups or types of language, the pseudo-labels it generates could reinforce these biases. The EXIST 2021 dataset does not include demographic information, such as age or gender, about the annotators, which might introduce biases into the data. This omission is largely due to the use of crowdsourcing for labeling and the need to preserve anonymity on these platforms. However, the EXIST 2022 dataset was labeled by expert annotators with balanced consideration of gender and age demographics. As observed in our results, both the EXIST 2021 and 2022 datasets demonstrate consistency across all techniques used, suggesting that, even if some biases may exist, the overall results are reliable and the generalization is robust. However, it is important to note that while the EXIST dataset includes data from two social networks and two languages, the model's ability to generalize to other platforms or languages may be limited. Furthermore, despite efforts during the development of the EXIST 2021 dataset to account for various potential sources of bias and ensure representativeness (Rodrguez-Sánchez et al. 2021, 2022), challenges remain. For instance, our analysis has shown that messages containing sarcasm and irony present significant difficulties for our system, making accurate classification more challenging in these cases.

While our current approach has demonstrated promising results in detecting and categorizing sexism on social media, there is still room for improvement. Sexist language can vary significantly across cultures and platforms, influenced by cultural norms, local idioms, and the nature of user interactions specific to each platform. Although our current model shows promising results in handling data from two major social networks and languages, extending this approach to other platforms and a wider range of languages could present new challenges and opportunities. Future research could investigate the nuances of sexist language across different cultural contexts and develop model adaptations to consider these variations.

Another potential direction for future work is to investigate the integration of sarcasm and irony detection into our system. As our manual error analysis has shown, a significant number of false negatives arise from tweets that contain sarcasm or irony, which our model struggles to classify correctly. The ability to detect sarcasm and irony could potentially lead to more accurate and robust classification of sexist content. To achieve this, we could explore the development of a dedicated sarcasm and irony detection module that could be incorporated into our system, allowing us to better handle the nuances of language and the subtleties of humor and irony in social media text.

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

[1] https://files.pushshift.io/gab/.

[2] https://www.mturk.com/.

[3] https://huggingface.co/sentence-transformers/stsb-xlm-r-multilingual.

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